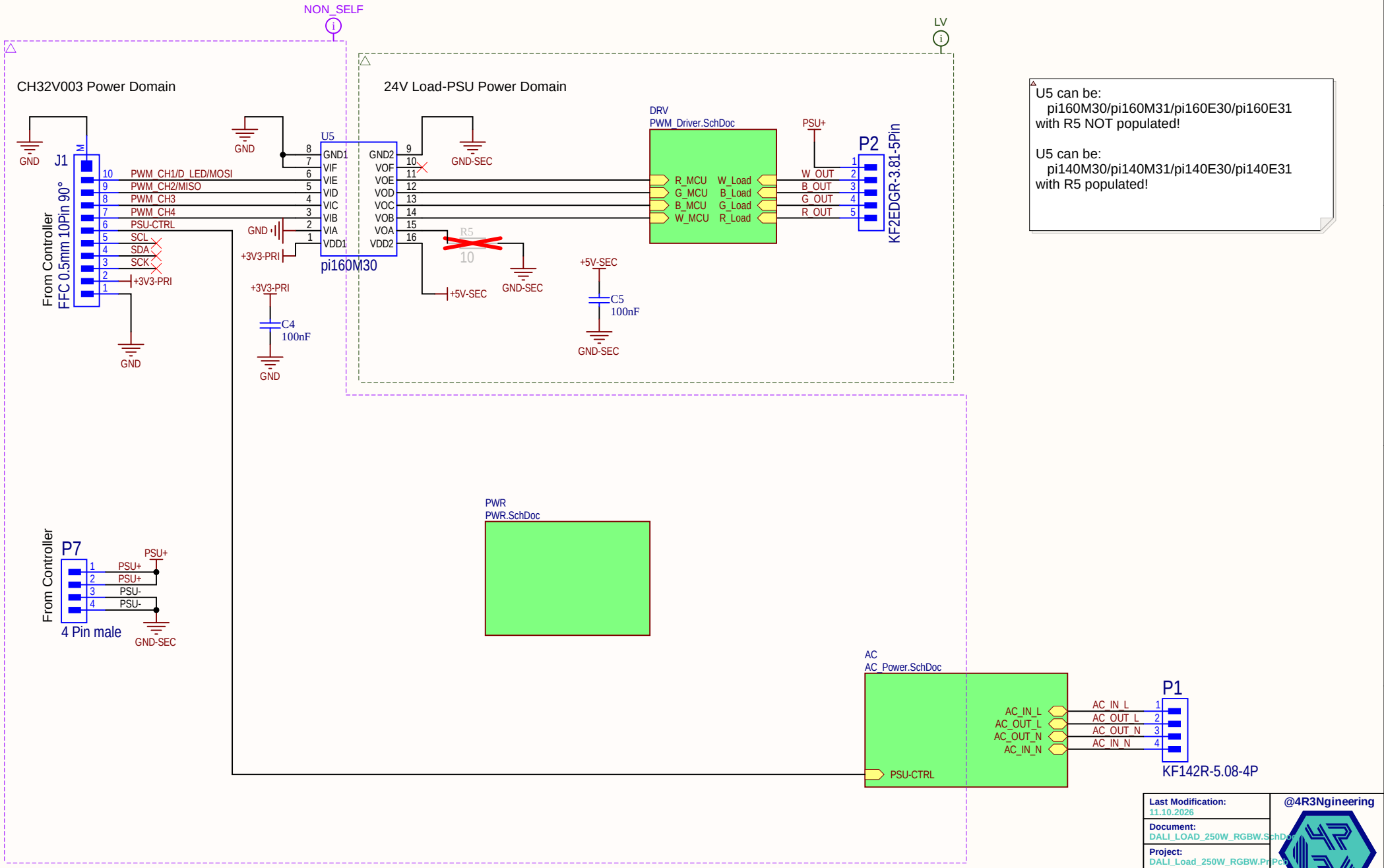
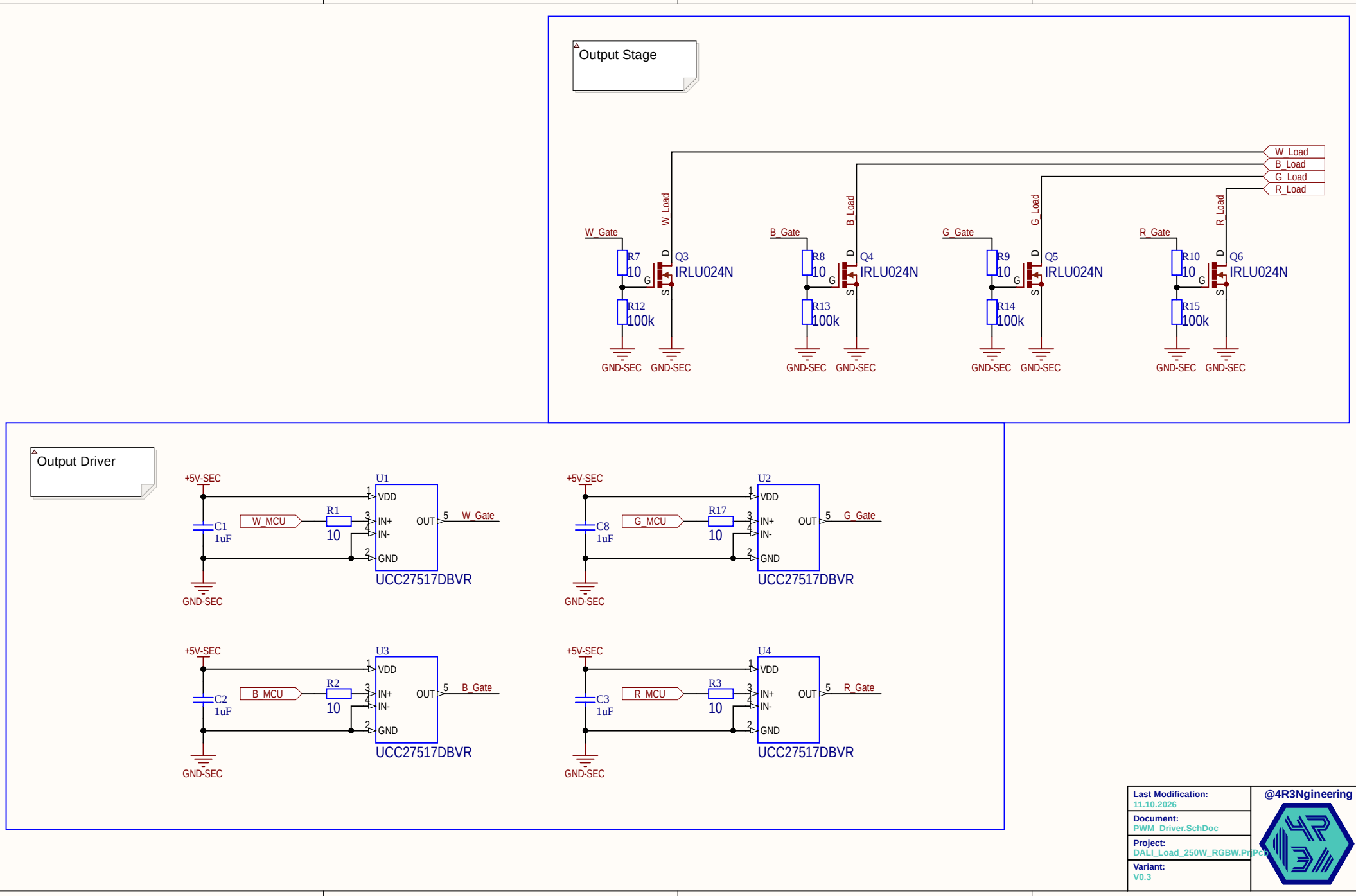
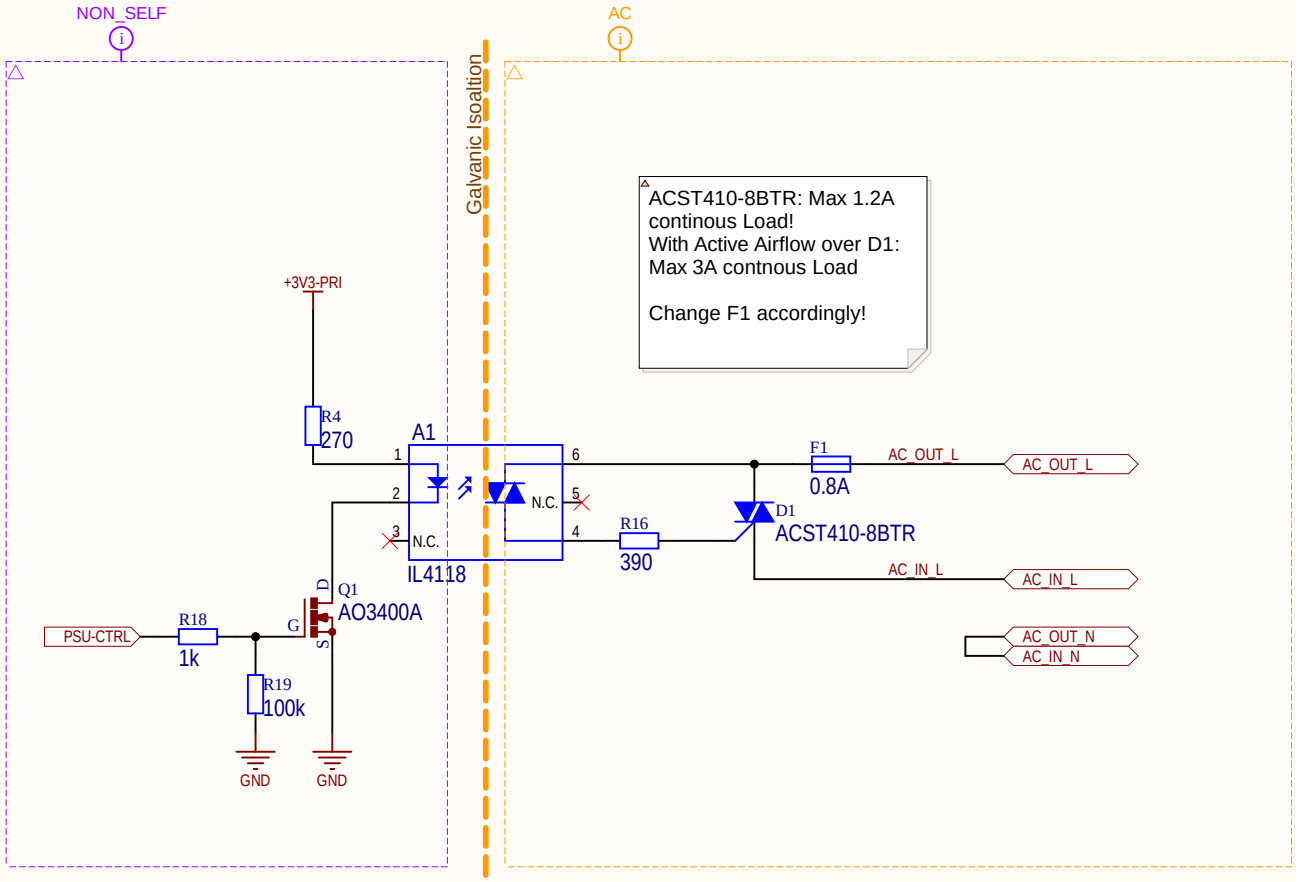
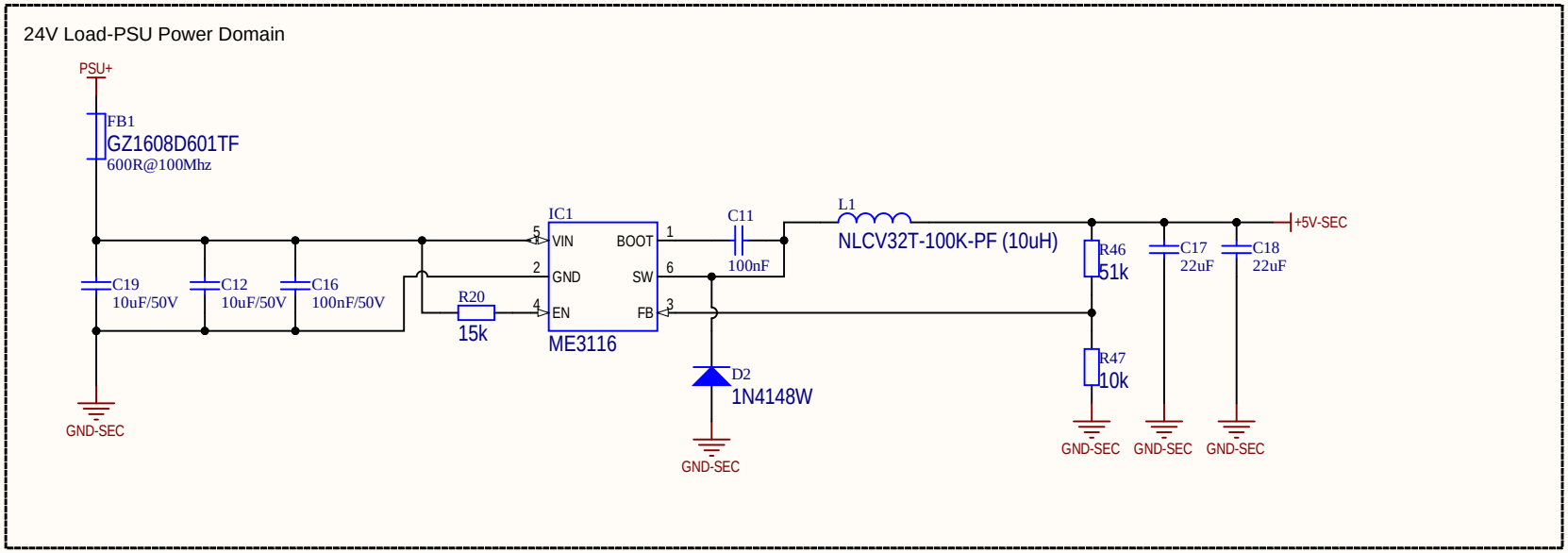

NON_SELF
LV
CH32V003 Power Domain
24V Load-PSU Power Domain
U5 can be:
pi160M30/pi160M31/pi160E30/pi160E31
with R5 NOT populated!
U5 can be:
pi140M30/pi140M31/pi140E30/pi140E31
with R5 populated!
J1
From Controller
FFC 0.5mm 10Pin 90°
PWM_CH1/D_LED/MOSI
PWM_CH2/MISO
PWM_CH3
PWM_CH4
PSU-CTRL
SCL
SDA
SCK
+3V3-PRI
GND
U5
pi160M30
GND1
VIF
VIE
VID
VIC
VIB
VIA
VDD1
GND2
VOF
VOE
VOD
VOC
VOB
VOA
VDD2
GND-SEC
+5V-SEC
R5
10
C4
100nF
C5
100nF
DRV
PWM_Driver.SchDoc
R_MCU
G_MCU
B_MCU
W_MCU
W_Load
B_Load
G_Load
R_Load
PSU+
P2
KF2EDGR-3.81-5Pin
W_OUT
B_OUT
G_OUT
R_OUT
P7
From Controller
4 Pin male
PSU+
PSU-
PWR
PWR.SchDoc
AC
AC_Power.SchDoc
AC_IN_L
AC_OUT_L
AC_OUT_N
AC_IN_N
PSU-CTRL
P1
KF142R-5.08-4P
Last Modification:
11.10.2026
Document:
DALI_LOAD_250W_RGBW.SchDoc
Project:
DALI_Load_250W_RGBW.PrjPcb
Variant:
V0.3
@4R3Ngineering


Output Stage
W_Load
B_Load
G_Load
R_Load
W_Gate
B_Gate
G_Gate
R_Gate
R7
10
Q3
IRLU024N
R8
10
Q4
IRLU024N
R9
10
Q5
IRLU024N
R10
10
Q6
IRLU024N
R12
100k
R13
100k
R14
100k
R15
100k
GND-SEC
Output Driver
+5V-SEC
U1
U2
U3
U4
VDD
IN+
IN-
GND
OUT
W_MCU
B_MCU
G_MCU
R_MCU
R1
R2
R3
R17
C1
C2
C3
C8
1uF
UCC27517DBVR
Last Modification:
11.10.2026
Document:
PWM_Driver.SchDoc
Project:
DALI_Load_250W_RGBW.PrjPcb
Variant:
V0.3
@4R3Ngineering



Last Modification:	11.10.2026
Document:	AC_Power.SchDoc
Project:	DALI_Load_250W_RGBW.PrjPcb
Variant:	V0.3

@4R3Ngineering



Last Modification:	11.10.2026
Document:	PWR.SchDoc
Project:	DALI_Load_250W_RGBW.PrjPcb
Variant:	V0.3

@4R3Ngineering